2747711595@qq.com: D260515002812

LETTER *Special Section on Cryptography and Information Security*

Addend Dependency of Differential/Linear Probability of Addition

Hiroshi MIYANO†, *Member*

SUMMARY This letter gives a study of addition $Y = X + K \bmod 2^w$ which is used in some cryptosystems as RC5. Our results enables us to express the differential and linear probability of addition as a function of addend K. To detect a good differential characteristics or linear approximation of a cryptosystem in which extended key is used as addend, we need to consider how the characteristics or approximations behave depending upon the value of the addend, which are clarified by our results.

key words: *common key cryptosystem, differential cryptanalysis, linear cryptanalysis, RC5, addition*

1. Introduction

To evaluate the strength of a cryptosystem, it is inevitable to estimate the efficiency of the differential cryptanalysis [1] and the linear cryptanalysis [2] over the cryptsystem. These efficiency depend on the best differential/linear probability. However, in some cases as RC5 [3], these probabilities depend on the key which is unknown but fixed during an attack, as Moriai et al. pointed out [5].

In such a case, we have to consider the key dependency of differential/linear probability of the cryptosystem to evaluate it. The cause of such a key dependency of RC5 is the addend dependency of differential/linear probability of addition. Generally, when we evaluate the strength of a cryptosystems using addition as a parts against differential/linear cryptanalysis, we have to study addend dependency of the probabilities of addition.

Let us see an example of addend dependency of linear approximation of addition. Let $Y = X+K \bmod 2^w$. And let X_i, K_i and Y_i be the i-th bits of X, K and Y respectively where the least significant bit is defined as the 0th bit. And we are to calculate $\Pr(X_1 \oplus K_1 \oplus Y_1 = 0)$.

It is clear that $X_1 \oplus K_1 \oplus Y_1 \neq 0$ if and only if $X_0 = K_0 = 1$. So, if we were on the assumption that X and K were random,

$$\Pr(X_1 \oplus K_1 \oplus Y_1 = 0) = \frac{3}{4}.$$

However, this assumption is improper. Remind that K is fixed during an attack. That mean that the phenomenon we observe is based on just one specific key. Therefore, we have to know what will happen for each specific value of the fixed key. In our example,

$$\Pr(X_1 \oplus K_1 \oplus Y_1 = 0) = \begin{cases} 1 & (K_0 = 0) \\ \frac{1}{2} & (K_0 = 1). \end{cases}$$

So we can express this situation as

$$\Pr(X_1 \oplus K_1 \oplus Y_1 = 0) = 1 - \frac{1}{2}K_0.$$

Note that the linear approximation probability is given as a function of K, not necessary a constant value.

Generally, linear approximation probability of a function $Y = f(X, K)$ is expressed as

$$LAP_f(\Gamma X, \Gamma K, \Gamma Y) = \Pr_x(\Gamma X \cdot X \oplus \Gamma K \cdot K \oplus \Gamma Y \cdot f(X, K) = 0)$$

where $\Gamma X, \Gamma K$ and ΓY are so-called *mask values* of X, K and Y respectively and the symbol of '$\cdot$' denotes the interior product of two vectors. Similarly, differential characteristic probability of function f is expressed as

$$DCP_f(\Delta X, \Delta Y) = \Pr_x(f(X, K) \oplus f(X \oplus \Delta X, K) = \Delta Y)$$

where ΔX and ΔY are fixed *differential values* of X and Y respectively. Obviously, LAP_f and DCP_f are functions of K. However, in some special case, the case where $f(X, K) = X \oplus K$ for example, factor of K are all eliminated, and as the result, LAP_f and/or DCP_f are/is constant(s).

In the traditional works applying differential or linear cryptanalysis, LAP_f's or DCP_f's are constants because the target cryptosystems have properties that the extended keys are captured into data with bitwise-xor, as DES or FEAL.

The aim of this paper is to clarify LAP and DCP of addition on modulo 2^w for a preparation to estimate the strength of cryptosystems against linear and differential cryptanalysis in which extended keys are used as addend in F-functions.

2. Preliminaries

Let us define some notations.

Manuscript received March/26/1997.
Manuscript revised July/15/1997.

†The author is with Yokohama Research Center, Telecommunications Advancement Organization of Japan, Yokohama-shi, 221 Japan.

2747711595@qq.com: D26051500281ZX00

The purpose of this paper is to formulate differential and linear probabilities of the addition $Y = X + K \bmod 2^w$. Let the i-th bit of X, K and Y be X_i, K_i and Y_i respectively where the 0th bit is the least significant bit.

Let us denote the carry bit up to the i-th bit as λ_i.

Suppose that V is a value of w bit and $0 \leqq t \leqq s \leqq w$. Let,

$$\mathrm{lsig}(V) = \max_{i \in [0,w]} i \mid \forall j < i \; V_j = 0,$$

$$\mathrm{msig}(V) = \min_{i \in [-1,w-1]} i \mid \forall j > i \; V_j = 0,$$

$$\mathrm{val}(V, s, t) = \frac{(V \bmod 2^s) - (V \bmod 2^t)}{2^s}.$$

For shortness, let us promise that the logical expression 'x' is *false/true* if a binary bit x is 0/1 respectively.

The differential and linear probability of addition DCP_+ and LAP_+ to be formulated is

$$DCP_+(\Delta X, \Delta Y) = \Pr_X((X + K) \oplus ((X \oplus \Delta X) + K) = \Delta Y),$$

$$LAP_+(\Gamma X, \Gamma K, \Gamma Y) = \Pr_X(\Gamma X \cdot X \oplus \Gamma K \cdot K \oplus \Gamma Y \cdot (X + K) = 0).$$

$\Delta X, \Delta Y$ are differential values of X, Y respectively, and $\Gamma X, \Gamma K$ and ΓY are mask value of X, K and Y respectively. For convenience, let us denote

$$\Delta F = (X + K) \oplus ((X \oplus \Delta X) + K).$$

So,

$$DCP_+(\Delta X, \Delta Y) = \Pr_X(\Delta F = \Delta Y).$$

Also, we simply denote $\Pr()$ instead of $\Pr_X()$.

In this paper, we discuss on the assumption that

$$\Gamma K = \Gamma X \vee \Gamma Y.$$

This assumption is not essential however.

Finally, we define $S(i), T(i)$ as follows.

$$S(i) = (X_i \wedge \Gamma X_i) \oplus (Y_i \wedge \Gamma Y_i) \oplus (K_i \wedge \Gamma K_i),$$

$$T(i) = \bigoplus_{k=0}^{i} S(k).$$

One of our purpose can be stated as to formulate $\Pr(\neg T(w))$ in other words.

3. Addend Dependency of DCP

This section shows addend dependency of differential probability of addition to formulate DCP for given ΔX and ΔY.

Using Theorems 2, 3, 4, 5, and corollary 1 recursively, we can formulate $DCP_+()$. Proofs are omitted.

Theorem 1: $\Delta F_0 = \Delta X_0$.

Theorem 2: $\Pr(\Delta F_i \neq \Delta X_i | \neg \Delta X_{i-1} \wedge \neg \Delta F_{i-1}) = 0$.

□

Theorem 3: For $0 < i \leqq w$,

$$\Pr(\Delta F_i \neq \Delta X_i | \neg \Delta X_{i-1} \wedge \Delta F_{i-1}) = \frac{1}{2}.$$

□

Corollary 1:

$$\forall i \leqq \mathrm{lsig}(\Delta X) \; \Delta F_i = \Delta X_i.$$

□

Theorem 4: For $0 < i \leqq w$,

$$\Pr(\Delta F_i \neq \Delta X_i | \Delta X_{i-1} \wedge \neg \Delta F_{i-1}) = \frac{1}{2}.$$

□

Before showing the Theorem 5, we show 2 lemmas which lead the theorem. Let,

$$T_1 = (\Delta X_{i-1} \wedge \Delta F_{i-1} \wedge (\forall j < i - 1 \; \neg \Delta X_j \wedge \neg \Delta F_j)),$$

$$T_2 = (\Delta X_{i-1} \wedge \Delta F_{i-1} \wedge (\exists j < i - 1 \text{ s.t. } (j < \forall h < i \; \neg \Delta X_h \wedge \neg \Delta F_h) \wedge (\Delta X_j \vee \Delta F_j))).$$

Lemma 1:

$$\Pr(\Delta F_i \neq \Delta X_i | T_1) = |K_{i-1} - \mathrm{val}(K, i - 1, 0)|.$$

□

Lemma 2:

$$\Pr(\Delta F_i \neq \Delta X_i | T_2) = |K_{i-1} - \mathrm{val}(K, i - 1, j + 1) + 2^{j-i+2} K_j|.$$

□

Theorem 5:

$$\Pr(\Delta Y_i \neq \Delta X_i) = |K_{i-1} - \mathrm{val}(K, i - 1, j + 1) + 2^{j-i+2} K_j|.$$

where, $j = \mathrm{msig}((\Delta X \wedge \Delta F) \bmod 2^{i-1})$, and $K_{-1} = 0$.

□

Now we have the following corollary.

Corollary 2: $DCP_+(\Delta X, \Delta Y) \neq 0$, iff

1. $0 \leqq \forall i \leqq \mathrm{lsig}(\Delta X), \Delta Y_i = \Delta X_i$,
2. $\Delta X_{i-1} = \Delta Y_{i-1} = 0 \Rightarrow \Delta Y_i = \Delta X_i$, and
3. If $\Delta X_{i-1} = 1$ or $\Delta Y_{i-1} = 1$, ΔY_i can be any of $\{0,1\}$.

□

Suppose that we are to formulate $DCP_+()$ for $\Delta X = 000100010000$, and $\Delta Y = 001101110000$. We can formulate the DCP using our results as follows. Proof is remained to the readers.

$$\mathrm{DCP}_+(000100010000, 001101110000) = \tfrac{1}{8} \cdot |K_8 - \tfrac{1}{2} K_7 + \tfrac{1}{2} K_6| \cdot |K_4 - \mathrm{val}(K, 4, 0)|.$$

2747711595@qq.com: D26051500281ZX0

4. Addend Dependency of LAP

This section provide a method to describe $\Pr(T(w))$ as a function of addend K.

The corollary 3 introduced by the theorems 6 and 7 clarifies the sufficient condition of mask values not to give a meaningful, not equal to $\frac{1}{2}$ in other words, linear approximation to reduce the mask values to be considered. To formulate $\Pr(T(w))$, check that the condition of the corollary 3 is not satisfied, apply the theorem either 8 or 9, and after applying theorem 9, apply the theorems 6, 10, 11, and 12 recursively. Most of the cases, it may be more convenient to apply the Corollary 4 thru 6 instead of the theorem 12.

To begin with, we show the theorem 6 and 7.

Theorem 6:

$$(\Gamma X_i \oplus \Gamma Y_i) \Rightarrow \Pr(T(i)) = \frac{1}{2}.$$

□

Theorem 7:

$$\Pr(T(w)) = \Pr(T(\max\{\mathrm{msig}(\Gamma X), \mathrm{msig}(\Gamma Y)\})).$$

□

These two theorems follow the following corollary which was already shown in [5] as it's theorem 2.

Corollary 3:

$$(\mathrm{msig}(\Gamma X) \neq \mathrm{msig}(\Gamma Y)) \Rightarrow \Pr(T(w)) = \frac{1}{2}.$$

□

For this corollary, to formulate $\Pr(T(w))$, we only have to discuss on the assumption of $\Gamma X_i \wedge \Gamma Y_i$. In the following Theorem 8 and 9, we assume $\Gamma X_i \wedge \Gamma Y_i$, that is, $S(i) = X_i \oplus Y_i \oplus K_i$.

Theorem 8: Suppose $\Gamma X_i \wedge \Gamma Y_i$, if $T(i) = S(i)$, then

$$\Pr(T(i)) = \mathrm{val}(K, i, 0).$$

□

Theorem 9: Suppose $\Gamma X_i \wedge \Gamma Y_i$, if $T(i) \neq S(i)$, then

$$\begin{aligned}&\Pr(T(i))\\ &\quad = \Pr(\lambda_i|\neg T(i-1))\Pr(\neg T(i-1))\\ &\qquad + \Pr(\neg\lambda_i|T(i-1))\Pr(T(i-1)).\end{aligned}$$

□

Theorems 8 and 9 provide a method to formulate $\Pr(T(i))$. However, we also need the way to calculate the right side of the equation of the theorem 9 for our purpose. To do that, it is sufficient to give a method to calculate $\Pr(\lambda_i|T(i-1))$ and $\Pr(\lambda_i|\neg T(i-1))$.

The following theorems describe the formula for $\Pr(\lambda_i|T(i)), \Pr(\lambda_i|\neg T(i))$ for each cases. Before showing these theorems, we sort out this partition of cases in a table.

	Th.	Condition of the case		
A	(8)	$\forall j < i\ \neg\Gamma X_j \wedge \neg\Gamma Y_j$		
B	10,11	ΓX_{i-1}	$\wedge\ \neg\Gamma Y_{i-1}$	$\wedge\ T3$
C	10,11	$\neg\Gamma X_{i-1}$	$\wedge\ \Gamma Y_{i-1}$	$\wedge\ T3$
D	10,11	ΓX_{i-1}	$\wedge\ \Gamma Y_{i-1}$	$\wedge\ T3$
E	12	$\neg\Gamma X_{i-1}$	$\wedge\ \neg\Gamma Y_{i-1}$	$\wedge\ T4$
F	12	ΓX_{i-1}	$\wedge\ \neg\Gamma Y_{i-1}$	$\wedge\ T4$
G	12	$\neg\Gamma X_{i-1}$	$\wedge\ \Gamma Y_{i-1}$	$\wedge\ T4$
H	12	ΓX_{i-1}	$\wedge\ \Gamma Y_{i-1}$	$\wedge\ T4$

Where,
$T3 = \forall j < i-1\ (\neg\Gamma X_j \wedge \neg\Gamma Y_j)$,
$T4 = \exists j < i-1\ (\Gamma X_j \vee \Gamma Y_j)$.

The following theorems can be proved in the similar way. So we only show the proof of theorem 10.

Theorem 10:

$$\Pr(\lambda_i|T(i-1)) = \begin{cases} \mathrm{val}(K, i-1, 0) & (\text{caseB})\\ K_{i-1} & (\text{caseC})\\ \frac{1}{2} + \frac{1}{2}K_{i-1} & (\text{caseD}).\end{cases}$$

□

Theorem 11:

$$\Pr(\lambda_i|\neg T(i-1)) = \begin{cases} K_{i-1} & (\text{caseB})\\ \mathrm{val}(K, i-1, 0) & (\text{caseC})\\ \frac{1}{2}K_{i-1} & (\text{caseD}).\end{cases}$$

□

Theorem 12: Let a be any of $\{0, 1\}$.

$$\begin{aligned}&\Pr(\lambda_i|T(i-1) = a)\\ &= \begin{cases} \frac{1}{2}K_{i-1} + \frac{1}{2}\Pr(\lambda_{i-1}|T(i-2) = a) & (\text{caseE, G})\\ \frac{1}{2}K_{i-1} + \frac{1}{2}\Pr(\lambda_{i-1}|T(i-2) = 1-a) & (\text{caseF})\\ \frac{1}{2}K_{i-1} + \frac{1}{2}\Pr(\lambda_{i-1}|T(i-2) = 1-a)\\ \quad + \frac{1}{2}K_{i-1}\{\Pr(\lambda_{i-1}|T(i-2) = 1-a)\\ \qquad - \Pr(\lambda_{i-1}|T(i-2) = a)\}. & (\text{caseH})\end{cases}\end{aligned}$$

□

Using the following corollaries instead of the theorem 12, we can formulate $\Pr(T(i))$ more easily.

Corollary 4:

$$\begin{aligned}&\mathrm{val}(K, i+h, i) + \frac{1}{2^h}\Pr(\lambda_i|T(i-1) = a)\\ &\quad = \mathrm{val}(K, i+h, i-1)\end{aligned}$$

2747711595@qq.com: D26051500281ZX00

$$+\begin{cases}\frac{1}{2^{h+1}}\Pr(\lambda_{i-1}|T(i-2)=a) & (\text{caseE, G})\\ \frac{1}{2^{h+1}}\Pr(\lambda_{i-1}|T(i-2)=1-a) & (\text{caseF})\\ \frac{1}{2^{h+1}}\Pr(\lambda_{i-1}|T(i-2)=1-a) & \\ +\frac{K_{i-1}}{2^{h+1}}\{\Pr(\lambda_{i-1}|T(i-2)=1-a) & \\ \quad -\Pr(\lambda_{i-1}|T(i-2)=a)\}. & (\text{caseH})\end{cases}$$

□

Corollary 5:

$$\begin{aligned}&\mathrm{val}(K,i+h,i)\\ &\quad+\frac{1}{2^{h+1}}\{\Pr(\lambda_i|T(i-1))+\Pr(\lambda_i|\neg T(i-1))\}\\ &=\mathrm{val}(K,i+h,i-1)\\ &\quad+\frac{1}{2^{h+2}}\{\Pr(\lambda_{i-1}|T(i-2))\\ &\quad+\Pr(\lambda_{i-1}|\neg T(i-2))\}\end{aligned}$$

□

Corollary 6:

$$\begin{aligned}&\Pr(\lambda_i|T(i-1))-\Pr(\lambda_i(\neg T(i-1)))\\ &=\begin{cases}\frac{1}{2}\{\Pr(\lambda_{i-1}|T(i-2)) & \\ \quad-\Pr(\lambda_{i-1}|\neg T(i-2))\} & (\text{caseE, G})\\ \frac{1}{2}\{\Pr(\lambda_{i-1}|\neg T(i-2)) & \\ \quad-\Pr(\lambda_{i-1}|T(i-2))\} & (\text{caseF})\\ \left(\frac{1}{2}+K_{i-1}\right)\{\Pr(\lambda_{i-1}|\neg T(i-2)) & \\ \quad-\Pr(\lambda_{i-1}|T(i-2))\}. & (\text{caseH})\end{cases}\end{aligned}$$

□

This section shows an example to formulate $LAP_+()$. In this case, the data length is 8.

Suppose, $\Gamma X = 01001101$, $\Gamma Y = 01101100$, $\Gamma K = 01101101$. In such a case, $LAP_+(\Gamma X, \Gamma Y, \Gamma K)$ is formulated as follows.

$$\begin{aligned}&LAP_+(\Gamma X,\Gamma Y,\Gamma K)\\ &\quad=\Pr(\neg T(7))\\ &\quad=1-\Pr(T(7))\\ &\quad=\frac{1}{2}-\frac{1}{16}\left(\frac{1}{2}+K_3\right)\left(\frac{1}{2}+K_2\right)K_0\end{aligned}$$

Proof is remained to the readers.

5. Conclusion

In this paper, we have proposed a method to formulate differential and linear probability of addition as a function of addend. Our result will be useful to detect differential characteristic or linear approximation of such a cryptosystem that uses addition to capture the extended key as RC5.

Moreover, another kind of differential or linear cryptanalysis may be possible using our result. When we analyze statistical data of many plain-encrypted text pairs with some differential characteristic or linear approximation, observed probability may indicate some addend dependency which could be a clue to find values of bits of extended key on which the probability depends. That is a further work.

Acknowledgement

The author thanks to the project leader Prof. Tsujii and sub-leaders Prof. Kaneko *et al.* of '*TAO reserach project on info-communication security*' for their helpful comments. And also thanks to researchers of the project who have often joined the discussion.

References

[1] E. Biham and A. Shamir, "Differential cryptanalysis of DES-like cryptosystems," Journal of Cryptology, vol.4, pp.3–72, 1991.
[2] M. Matsui, "Linear cryptanalysis method for DES cipher," EUROCRYPT'93 Extended Abstracts, 1993.
[3] R.L. Rivest, "The RC5 encryption algorithm," FAST SOFTWARE ENCRYPTION, 2nd International Workshop, Springer-Verlag, 1995.
[4] B.S. Kaliski and Y.L. Yin, "On Differential and Linear Cryptanalysis of the RC5 Encryption Algorithm," Advances in Cryptology — CRYPTO'95, Springer-Verlag, 1995.
[5] S. Moriai, K. Aoki, and K. Ohta "On the linear probability of RC5 (in Japanese)," Proc. 1996 Symposium of Cryptography and Information Security, SCIS96-12C, 1996.
[6] H. Miyano, "Differential/linear probability of addition upon the addend dependency (in Japanese)," Workshop on Design and Evaluation of Cryptographic Algorithm, Nov. 1996.